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BTECH
(SEM III) THEORY EXAMINATION 2025-26
DISCRETE STRUCTURES & THEORY OF LOGIC

TIME: 3 HRS**M.MARKS: 70**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A**1. Attempt all questions in brief.****2 x 07 = 14**

Q no.	Question	CO
a.	Define a lattice and state the meaning of upper bound and lower bound in a lattice.	1
b.	What is meant by a reflexive relation? State its basic property.	2
c.	Find the domain and range of the function $f(x)=1- x $.	2
d.	What is meant by an existential quantifier? Explain it with a simple statement.	3
e.	Define zero divisor of a ring.	4
f.	How many 4 digits number can be formed by using the digits 2,4,6,8 when the repetition of digits is allowed.	5
g.	Draw a graph that has a Hamiltonian path not have a Hamiltonian circuit.	5

SECTION B**2. Attempt any three of the following:****07 x 3 = 07**

Q no.	Question	CO
a.	If R is an equivalence relation on A, then prove that R^{-1} is also equivalence relation on A.	1
b.	Explain the basic concepts of Boolean algebra and state any four fundamental laws of Boolean algebra.	2
c.	Translate the following statements in symbolic form: (i) The sum of two positive integer is always positive. (ii) Everyone is loved by someone. (iii) Some people are not admired by everyone. (iv) If a person is female and is a parent, then this person is someone's mother.	3
d.	State and prove Lagrange's theorem.	4
e.	Show that in any room of people who have been doing handshaking there will always be at least two people who have shaken hands the same number of times.	5

SECTION C**3. Attempt any one part of the following:****07 x 1 = 07**

Q no.	Question	CO
a.	Define Lattice and the properties of lattice.	1
b.	Explain the following terms: (i) POSET (ii) Hasse Diagram	1



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4. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO
a.	Use a Karnaugh map to find a minimal sum for $E = y't' + y'z't + x'y'zt + yzt'$	2
b.	Show that the number of minimal Boolean function in n variables are 2^n .	2

5. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO
a.	Prove that the statement “if n is an integer, then $n^2 - n + 4!$ Is a prime number” is false.	3
b.	Explain the following terms with suitable example: (i) Converse (ii) Disjunction (iii) Conjunction	3

6. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO
a.	Let H be a subgroup of a finite group G . Prove that order of H is a divisor of order G .	4
b.	How many generators are there of the cyclic group G of order 10.	4

7. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO
a.	Let G be a 3-regular graph with n vertices. What is the sum of the degree of the vertices? Show that in such a graph n must be even.	5
b.	Prove that if a connected graph G is decomposed into two subgraphs g_1 and g_2 , there must be at least one vertex common between g_1 and g_2 .	5



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Note: सभी अनुभागों का प्रयास करें। यदि कोई डेटा गायब हो तो उपयुक्त विकल्प चुनें।

SECTION A

1. सभी प्रश्नों का संक्षेप में उत्तर दें।

2 x 07 = 14

Q no.	Question	CO	Level
a.	Let $A = \{x: x \text{ is a prime number less than } 20\}$ and $B = \{x: x \text{ is an odd number less than } 20\}$. Compute $A \cup B$ and $A \cap B$. मान लें कि $A = \{x: x \text{ एक अभाज्य संख्या है जो } 20 \text{ से कम है}\}$ और $B = \{x: x \text{ एक विषम संख्या है जो } 20 \text{ से कम है}\}$ $A \cup B$ और $A \cap B$ की गणना करें।	1	K2
b.	Let $A = \{1, 2, 3\}$ and $B = \{a, b\}$. Compute the total number of possible relations from A to B . मान लें कि $A = \{1, 2, 3\}$ और $B = \{a, b\}$, A से B तक संभावित संबंधों की कुल संख्या की गणना करें।	1	K2
c.	If $P(x)$ represents " x is a prime number" and $Q(x)$ represents " x is odd," write the following statements in predicate logic: (a) "There exists an even prime number." (b) "All prime numbers greater than 2 are odd." यदि $P(x)$ " x एक अभाज्य संख्या है" को प्रदर्शित करता है और $Q(x)$ " x विषम है" को प्रदर्शित करता है, तो निम्नलिखित कथनों को गणक तर्क (predicate logic) में इस प्रकार लिखा जा सकता है: (a) "एक सम अभाज्य संख्या अस्तित्व में है।" (b) "सभी अभाज्य संख्याएँ जो 2 से बड़ी हैं, वे विषम होती हैं।"	2	K2
d.	Calculate the composite mapping $g \circ f$ if $f: R \rightarrow R$ is given by $f(x) = e^x$ and $g: R \rightarrow R$ is given by $g(x) = \sin x$. संयुक्त फलन $g \circ f$ की गणना करें यदि $f: R \rightarrow R$ इस प्रकार दिया गया है: $f(x) = e^x$ और $g: R \rightarrow R$ इस प्रकार दिया गया है: $g(x) = \sin x$ ।	2	K3
e.	Define Abelian group. एबेलियन समूह परिभाषित करें।	3	K1
f.	Solve in how many ways can you arrange the letters in the word "DISCRETE"? हल करें: शब्द "DISCRETE" के अक्षरों को कितने तरीकों से व्यवस्थित किया जा सकता है?	4	K3
g.	Illustrate the following graph using an adjacency matrix: A graph with vertices $V = \{A, B, C, D\}$ and edges $E = \{(A, B), (B, C), (C, D), (D, A)\}$. निम्नलिखित ग्राफ को एक आसन्नता मैट्रिक्स (Adjacency Matrix) का उपयोग करके चित्रित करें: एक ग्राफ जिसमें शीर्षकों (vertices) का समुच्चय $V = \{A, B, C, D\}$ और किनारों (edges) का समुच्चय $E = \{(A, B), (B, C), (C, D), (D, A)\}$ हो।	5	K3

SECTION B

2. निम्नलिखित में से किसी भी तीन का प्रयास करें:

07 x 3 = 21

a.	Examine $R = \{(a, b) \mid a \equiv b \pmod{m}\}$ is an equivalence relation on Z . Also ensure that if $x_1 \equiv y_1$ and $x_2 \equiv y_2$ then $(x_1 + x_2) \equiv (y_1 + y_2)$.	1	K3
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	जाँच करें कि $R = \{(a, b) \mid a \equiv b \pmod{m}\}$ एकसमानता संबंध है $\mathbb{Z}$ पर। इसके अलावा, यह भी सुनिश्चित करें कि यदि $x_1 \equiv y_1$ और $x_2 \equiv y_2$ तो $(x_1 + x_2) \equiv (y_1 + y_2)$ होगा।		
b.	Solve the following Boolean function using K-map: $F(A, B, C) = (1, 2, 5, 7)$ and $D(0, 4, 6)$ using SOP. निम्नलिखित बूलीय फलन को Karnaugh मैप (K-map) का उपयोग करके हल करें: $F(A, B, C) = (1, 2, 5, 7)$ और डॉट केयर शर्तें: $D(0, 4, 6)$ SOP रूप में हल करें।	2	K3
c.	Analyse the argument's validity: Premises: If a person is happy, they smile. John is smiling. Conclusion: John is happy. तर्क की वैधता का विश्लेषण करें: पूर्ववाक्य: यदि कोई व्यक्ति खुश है, तो वह मुस्कुराता है। जॉन मुस्कुरा रहा है। निष्कर्ष: जॉन खुश है।	3	K4
d.	In the group $\mathbb{Z}_{12}$ under addition modulo 12, consider the subgroup $H = \{0, 4, 8\}$. Compute all the distinct Cosets of H . समूह $\mathbb{Z}_{12}$ में, योग (addition) गुणनखंड 12 के तहत, उपसमूह $H = \{0, 4, 8\}$ पर विचार करें। सभी विभिन्न कोसेट (Cosets) की गणना करें।	4	K3
e.	Examine whether the graphs K_3 and a graph formed by adding a single vertex in the middle of one edge of K_3 are homeomorphic or not. यह जाँचें कि क्या ग्राफ K_3 और वह ग्राफ, जो K_3 की एक भुजा के मध्य में एक नया शीर्ष जोड़कर बनाया गया है, होमियोमोर्फिक (homeomorphic) हैं या नहीं।	5	K4

SECTION C

3. निम्नलिखित में से किसी एक भाग का प्रयास करें:

07 x 1 = 07

a.	Let $R = \{(1, 2), (2, 3), (3, 1)\}$ defined on $A = \{1, 2, 3\}$. Calculate the transitive closure of R using Warshall's algorithm. मान लें कि $R = \{(1, 2), (2, 3), (3, 1)\}$ समुच्चय $A = \{1, 2, 3\}$ पर परिभाषित है। Warshall के एल्गोरिदम का उपयोग करके R का संक्रमणशील आवरण (transitive closure) गणना करें।	1	K3
b.	i) Justify that $(\mathbf{D}_{42}, \vee)$ is lattice. ii) Let L_1 be the lattice defined as $\mathbf{D}_6$ and L_2 be the lattice $(\mathbf{P}(S), \leq)$, where $\mathbf{P}(S)$ be the power set defined on set $S = \{a, b\}$. Justify that the two lattices are isomorphic. i) यह प्रमाणित करें कि $(\mathbf{D}_{42}, \vee)$ एक लाटिस (lattice) है। ii) L_1 को $\mathbf{D}_6$ के रूप में परिभाषित किया गया है और L_2 को $(\mathbf{P}(S), \leq)$ के रूप में परिभाषित किया गया है, जहाँ $\mathbf{P}(S)$ सेट $S = \{a, b\}$ पर परिभाषित	1	K3



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	शक्ति समुच्चय (power set) है। यह प्रमाणित करें कि ये दोनों लाटिस आपस में समरूप (isomorphic) हैं।		
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4. निम्नलिखित में से किसी एक भाग का प्रयास करें: 07 x 1 = 07

a.	Solve the following Boolean functions using K-map: (i) $F(A, B, C, D) = \sum(m_0, m_1, m_2, m_4, m_5, m_6, m_8, m_9, m_{12}, m_{13}, m_{14})$ (ii) $F(A, B, C, D) = \sum(0, 2, 5, 7, 8, 10, 13, 15)$ निम्नलिखित बूलीय फलनों को K-map का उपयोग करके हल करें: (i) $F(A, B, C, D) = \sum(m_0, m_1, m_2, m_4, m_5, m_6, m_8, m_9, m_{12}, m_{13}, m_{14})$ (ii) $F(A, B, C, D) = \sum(0, 2, 5, 7, 8, 10, 13, 15)$	2	K3
b.	Let $f(x) = 3x + 5$ and $g(x) = x^2 - 2x + 1$. Evaluate $f(3)$ and Classify $f(x)$ and $g(x)$ as one-to-one, onto, or neither. मान लें कि $f(x) = 3x + 5$ और $g(x) = x^2 - 2x + 1$ । $f(3)$ का मूल्य निकालें और $f(x)$ और $g(x)$ को one-to-one, onto, या इनमें से कोई भी वर्गीकृत करें।	2	K3

5. निम्नलिखित में से किसी एक भाग का प्रयास करें: 07 x 1 = 07

a.	Justify that the following premises are inconsistent. (i) If Alex misses many classes through illness then he fails high school. (ii) If Alex fails high school, then he is uneducated. (iii) If Alex reads a lot of books then he is not uneducated. (iv) Alex misses many classes through illness and reads a lot of books. यह प्रमाणित करें कि निम्नलिखित पूर्ववाक्य असंगत हैं: (i) यदि एलेक्स कई कक्षाएँ बीमारी के कारण छोड़ता है, तो वह हाई स्कूल में फेल हो जाता है। (ii) यदि एलेक्स हाई स्कूल में फेल हो जाता है, तो वह अप्रशिक्षित हो जाता है। (iii) यदि एलेक्स बहुत सारी किताबें पढ़ता है, तो वह अप्रशिक्षित नहीं होता है। (iv) एलेक्स कई कक्षाएँ बीमारी के कारण छोड़ता है और बहुत सारी किताबें पढ़ता है।	3	K3
b.	Show the validity of the following argument: Hypotheses: "It is not sunny this afternoon and it is colder than yesterday. We will go swimming only if it is sunny. If we do not go swimming, then we will take a canoe trip. If we take a canoe trip, then we will be home by sunset." Conclusion: "We will be home by sunset." निम्नलिखित तर्क की वैधता को प्रदर्शित करें: परिकल्पनाएँ: "आज दोपहर धूप नहीं है और यह कल से ठंडा है। हम तैरने तभी जाएंगे जब धूप होगी। यदि हम तैरने नहीं जाते हैं, तो हम कैनो यात्रा पर जाएंगे। यदि हम कैनो यात्रा पर जाते हैं, तो हम सूर्यास्त तक घर वापस आ जाएंगे।" निष्कर्ष: "हम सूर्यास्त तक घर वापस आ जाएंगे।"	3	K3



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6. निम्नलिखित में से किसी एक भाग का प्रयास करें:

07 x 1 = 07

a.	Calculate the number of generators of the cyclic group of order 8. आदेश 8 के चक्रीय समूह के उत्प्रेरकों की संख्या की गणना करें।	4	K3
b.	Determine whether the set $H = \{0, 1, 5\}$ is a subgroup of $\mathbb{Z}_6$ under addition modulo 6. निर्धारित करें कि सेट $H = \{0, 1, 5\}$ $\mathbb{Z}_6$ के तहत 6 के modulo योग में उपसमूह है या नहीं।	4	K3

7. निम्नलिखित में से किसी एक भाग का प्रयास करें:

07 x 1 = 07

a.	Express the following (i) Euler graph and Hamiltonian graph (ii) Chromatic number of a graph (iii) Walk and path (iv) Bipartite graph निम्नलिखित को व्यक्त करें: (i) यूरलर ग्राफ़ और हैमिल्टनियन ग्राफ़ (ii) एक ग्राफ़ का क्रोमैटिक संख्या (iii) वॉक और पथ (iv) द्विविभाज्य ग्राफ़	5	K3
b.	Prove that in any group of 13 people, at least two must have their birthdays in the same month. यह प्रमाणित करें कि 13 लोगों के किसी भी समूह में, कम से कम दो लोगों का जन्मदिन एक ही महीने में होना चाहिए।	5	K3



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TIME: 3HRS

M.MARKS: 70

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2 x 7 = 14

Q no.	Question	Marks	CO
a.	Determine the greatest lower bound and least upper bound of the set $\{2, 3, 6\}$, if they exist, in the Poset $(D_{24}, /)$.	2	1
b.	Express power set of each of these sets. 1) $\{\emptyset, \{\emptyset\}\}$ 2) $\{a, \{a\}\}$	2	1
c.	Investigate whether the function $f(x) = x^2 - 1$ is injective or not for $f: \mathbb{R} \rightarrow \mathbb{R}$.	2	2
d.	Express $E(x, y, z) = xy + y'z$ into its complete sum-of-products form.	2	2
e.	Construct inverse of the following statement "If I wake up early in the morning, then I will be healthy."	2	3
f.	Show that identity element is unique in a group.	2	4
g.	Compare Euler circuit and Hamiltonian circuit.	2	5

SECTION B

2. Attempt any three of the following:

7 x 3 = 21

Q no.	Question	Marks	CO
a.	Construct the Hasse Diagram for $(P(S), \subseteq)$ where $P(S)$ is a power set defined on set $S = \{a, b, c\}$. Determine whether it is a Lattice or not.	7	1
b.	Solve the following Boolean functions using K-map: (i) $F(A, B, C, D) = \sum (m_0, m_1, m_2, m_4, m_5, m_6, m_8, m_9, m_{12}, m_{13}, m_{14})$ (ii) $F(A, B, C, D) = \sum (0, 2, 5, 7, 8, 10, 13, 15)$	7	2
c.	Show the validity of the following argument: hypotheses: "It is not sunny this afternoon and it is colder than yesterday. We will go swimming only if it is sunny. If we do not go swimming, then we will take a canoe trip. If we take a canoe trip, then we will be home by sunset." conclusion: "We will be home by sunset."	7	3
d.	Let $G = \{1, -1, i, -i\}$ with the binary operation multiplication be an algebraic structure, where $i = \sqrt{-1}$ then determine whether G is an Abelian group. Also if G is cyclic Group, then determine the generator of G .	7	4
e.	Explain Pigeon hole principle. Describe generalized form of Pigeon hole principle. If 6 colors are to paint 37 homes. Show that at least 7 of them will be of same color.	7	5



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SECTION C

3. Attempt any one part of the following:

7 x 1 = 7

Q no.	Question	Marks	CO
a.	Let R be a binary relation on the set of all strings of 0 and 1 such that $R = \{(a,b): a \text{ and } b \text{ have same number of 0's}\}$. Show that whether R is reflexive, symmetric, transitive or a partial order relation.	7	1
b.	Show that $(D_{42}, /)$ is lattice. Compare the distributive and complemented lattice with example.	7	1

4. Attempt any one part of the following:

7 x 1 = 7

Q no.	Question	Marks	CO
a.	Solve the following Boolean function using K-map: $F(A,B,C) = (1,2,5,7)$ and $D(0,4,6)$ using SOP.	7	2
b.	If $f: R \rightarrow R$, $g: R \rightarrow R$ and $h: R \rightarrow R$ defined by $f(x) = 3x^2+2$, $g(x) = 7x-5$ and $h(x) = 1/x$. Compute the following composition functions. (i) $(f \circ g \circ h)(x)$ (ii) $(g \circ g)(x)$ (iii) $(g \circ h)(x)$	7	2

5. Attempt any one part of the following:

7 x 1 = 7

Q no.	Question	Marks	CO
a.	Test the validity of the following argument. “If there was a ball game, then traveling was difficult. If they arrived on time, then traveling was not difficult. They arrived on time. Therefore, There was no ball game.”	7	3
b.	Describe $\exists$ and $\forall$ Quantifiers with example. “There is someone who got an A in the course” convert this sentence into predicate logic using quantifiers. Prove the following argument. All man are mortal. Socrates is a man. Therefore, Socrates is mortal.	7	3

6. Attempt any one part of the following:

7 x 1 = 7

Q no.	Question	Marks	CO
a.	Describe Algebraic structure, semigroup, monoid and group. Also explain the relationship among them.	7	4
b.	Consider group $G = \{1, 2, 3, 4, 5, 6\}$ under multiplication modulo 7. (a) Construct the multiplication table of G. (b) Compute 2^{-1} , 3^{-1} , 6^{-1} (c) Compute the orders and subgroups generated by 2 and 3. (d) Is G cyclic?	7	4

7. Attempt any one part of the following:

7 x 1 = 7

Q no.	Question	Marks	CO
a.	Compare bipartite and complete graph with example. Draw $K_{3,4}$ and K_5 . Explain why these two graphs are not planar.	7	5
b.	Show that $K_{3,3}$ satisfies in equality $ E \leq 3 V - 6$, but it is non-planar. (V =No. of Vertices, E =No. of Edges, R =No. of Regions)	7	5